THE OPEN PROBLEMS OF GROTHENDIECK'S 1973 BUFFALO LECTURES, AND WHAT BECAME OF THEM

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Abstract. In the spring and summer of 1973 Grothendieck gave three courses at the State University of New York at Buffalo, on topos theory, on algebraic geometry and on group schemes, whose recordings are online. In them he asks some twenty questions he cannot answer, states conjectures, and hands problems to the audience. We list them from the recordings, with the second at which each is asked, and say what the literature has made of each. Most are settled. Several went against what he expected: his characterisation of presheaf topoi, the non-existence of a right adjoint $f^!$, the colimits of accessible categories, and the descent of immersions that are not quasi-compact. Perrin proved his main conjecture on group schemes in 1976, one question is settled only under a large-cardinal axiom or an extra hypothesis, and the effectivity of finite equivalence relations is answered in a companion note. What still looks open, as he phrased it, concerns mostly algebraic theories over a fixed base.

1. The lectures and the method

The recordings are on a YouTube channel under the pseudonym « Erwin Schulhoff »: four videos for the topos course (April–July 1973), four for algebraic geometry (May–July) and two for group schemes (June–July), the last two also carrying some sessions of the algebraic geometry course. The edition has made an index of results for each course, with the second at which each statement is made, and reconstructed the blackboards [27]. For this note we searched the transcripts for the places where he says that something is not known, that he cannot prove it, that he expects or suspects it, or where he calls it a conjecture or a problem; we kept the ones he frames as questions, and dropped asides and routine omissions. Each problem below gives the video and the time (a link to that second), how he frames it, and its status, with the references we checked. Where we give an argument of our own, not found in the literature, it is marked $^{(e)}$.

The transcripts are first passes from captions and Whisper that no one has checked, and a few exchanges with the room are hard to attribute. He also corrected himself: on 10 July he retracted several claims of 2 July on algebraic theories (the transitivity of strict cogeneration, a two-step construction, a criterion by strict cogenerators), recorded in the edition's list of corrections [8]. We report the question as finally posed.

2. Topoi

Video 2 of the topos course is `PqCQM3Pcl6Q` unless stated.

Problem 1 (1:26:22 and 7:31:23)**.** *Are the presheaf topoi $\widehat{C}$ exactly the topoi that share all the exactness properties of* **Set***, including those involving infinite limits: arbitrary sums and products distributive, and arbitrary products of epimorphisms epimorphic?*

He suspects it, says it should not be too hard, and returns to it later as a characterisation of what he calls essential topoi, at least when they have generators.

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Status. False as stated. Presheaf topoi are characterised by a generating set of indecomposable projective objects (Bunge [5]). The right class for his two conditions is larger: Lucyshyn-Wright proved that the totally distributive categories with a set of generators are exactly the essential subtopoi of presheaf topoi [19, 1.5.9], and both conditions pass to any essential subtopos $s\colon E \to \widehat{C}$, since s^* preserves products, sums and epimorphisms and $s^*s_! \cong 1 \cong s^*s_*$[(e)]. A topos outside the presheaf topoi is the topos of sheaves on $[0,1]$ with the Scott topology, whose open sets are $[0,1]$ and the intervals $(a,1]$. It is a retract of the presheaf topos $[[0,1], \mathbf{Set}]$, the sheaves for the topology of all upper sets: the inclusion of the Scott opens among the upper sets and the map $U \mapsto \bigcup_{a\in U}\{x : x > a \text{ or } a = 0\}$ are frame homomorphisms whose composite is the identity[(e)]. So it is quasi-injective in the sense of Johnstone and Joyal [15] (a retract of a presheaf topos, as recalled in [19, §1.4]), hence totally distributive [19, 1.5.4], and it satisfies both conditions. It is not a presheaf topos: the subterminal objects of $\widehat{C}$ form a lattice of down-sets, in which every element is a join of completely join-prime elements, while among the Scott opens only $[0,1]$ is completely join-prime, $(a,1]$ being the union of the $(b,1]$ for $b > a$[(e)]. His « essential » is not the essential geometric morphism of later usage. Whether his two conditions alone are equivalent to total distributivity among Grothendieck topoi we have not found stated.

Problem 2 (4:05:50)**.** *For a morphism of topoi f, can f_* have a right adjoint $f^!$ on sheaves of sets without f being an equivalence?*

He has « a feeling » that it cannot, and concludes that the notation $f^!$ has no reason to exist for topoi.

Status. False, even for an open dense immersion, the case he was testing. For the open point j of the Sierpiński space S, whose sheaves are the maps $A \to B$ (sections over S, then over the open point), j_* is the diagonal $A \mapsto (A \to A)$, which is the constant-sheaf functor of S, and its right adjoint is the global-sections functor Γ[(e)]. More generally, for a small category C with a terminal object the global-sections functor $\Gamma\colon \widehat{C} \to \mathbf{Set}$, the direct image of $\widehat{C} \to \mathbf{Set}$, has a right adjoint: $\widehat{C}$ is a local topos, and such morphisms are the local maps studied by Johnstone and Moerdijk [16].

Problem 3 (5:18:59)**.** *Every type of algebraic structure, in a broad sense, should have a classifying topos with a universal model, added axioms should correspond to subtopoi, and the homotopy type of the classifying topos should be a classifying space of the structure.*

He states it as a programme and says he has checked it on a dozen or two examples.

Status. Carried out for geometric theories: every geometric theory has a classifying topos with a generic model (Joyal–Reyes; Makkai–Reyes [21]; Hakim for rings and local rings [13]), quotient theories correspond to subtopoi (Caramello [6]), and for groups and categories the classifying topos has the weak homotopy type of the classifying space (Moerdijk [23]). The restriction is that « structure » means geometric theory.

Problem 4 (8:38:47, 8:46:26)**.** *Can every exactness property be written as « every D-functor $I \to C$ factors through a given D-functor $I \to I'$ »? Is « every semigroup object is a monoid » an exactness property?*

This comes from a discussion with the room after the session, and who says what is not always clear.

Status. Formalised by Jacqmin and Janelidze as exactness sequents of sketches, which cover what they call first-order exactness properties, among them properties of internal structures such as an internal monoid being a group, and not higher-order ones such as having enough projectives [14]. « Every semigroup object is a monoid » holds only in the trivial category: the left-zero multiplication $X \times X \to X$ is associative, and a unit $e\colon 1 \to X$ for it forces $\mathrm{id}_X = e \circ !$, so $X \cong 1$; since every object carries this multiplication, the category is trivial[(e)].

Two further questions are answered on the spot. Asked for a criterion characterising sober spaces (1:55:37), he says he knows none that can be read off the topos, since a space and its sobrification have the same sheaves; the intrinsic criteria are standard [17]. And the universal problem for torsors under a group G of a topos S (`ORG14LNb5gE`, 3:15:58) he solves at once with the topos of G-objects of S, as in Giraud [12].

3. Algebraic theories relative to a class of limits

For a class D of limits and colimits, he considers the 2-category $\mathbf{Cat}_D$ of categories with D-limits and D-functors, and calls a theory algebraic if it is obtained from a set I of base objects by stepwise rules: the free D-envelope, adjoining arrows, imposing axioms.

Problem 5 (8:34:57)**.** *Are the algebraic theories relative to D exactly the theories $C \mapsto \mathrm{Hom}_D(R, C)$ 2-represented by objects R of $\mathbf{Cat}_D$?*

Status. Settled: yes for small R, which is his own argument and the modern presentation of small categories with limits by generators and relations (sketches, essentially algebraic theories [2, ch. 3]); no in general. For D the small limits, $R = \mathbf{Set}$ is representable, with $\mathbf{Set}^{\mathrm{op}}$ as its category of models in $\mathbf{Set}$, which is not accessible, while stepwise theories have locally presentable categories of models; he gives this example himself on 10 July. For D all small limits the algebraic R are, up to duality, the opposites of locally presentable categories, the form Gabriel–Ulmer duality gives [11].

Problem 6 (10:03:06)**.** *For which D are the envelope problem (the free D-completion of a category) and the localisation problem in $\mathbf{Cat}_D$ solvable? He knows them for finite limits and for topoi, and not for additive, abelian or Karoubian categories.*

Status. Settled for every small class D by Kelly's transfinite constructions and two-dimensional monad theory [18, 4], with explicit answers: the idempotent completion for Karoubian categories, the biproduct completion of the free preadditive category for additive ones, the free abelian category of Freyd and Adelman [10, 1], the exact completion of Carboni and Celia Magno [7]. For a large class, such as all small limits, the answers depend on size, as he noted on 10 July: localisations of large categories need not be locally small.

Problem 7 (`4gynb9uTEE4`, 2:03:54 and 2:10:23)**.** *« Problem three »: is $I \to R$ basic if and only if every strictly full subcategory of R stable under D-limits and containing the image of I is R itself? And, conjecturally, can I be enlarged so that no arrows remain to be adjoined, only axioms?*

Status. The direct implication holds; the converse fails for D all small limits, with $R = \mathbf{Set}$ and $I = \{2\}$, since every set is an equaliser of maps between powers of 2 but $\mathbf{Set}$ is not algebraic$^{(\mathrm{e})}$; for essentially small D the equivalence holds, everything being small. The second question has a positive answer in the locally presentable case by Gabriel–Ulmer: taking for I the opposite of the λ-presentable objects, the category of models is the category of λ-continuous functors on I, so that only axioms remain to be imposed [11], [2, 1.46]. For a fixed base I the question « basic if and only if strong generator » seems open as he phrased it, and so does the length of the closure.

Around these he asks whether faithfulness of $\mathrm{Hom}_D(R, C) \to C^I$ for all C suffices for algebraicity (9:40:36, 10:00:19). It does not, as he reports himself later, citing an example he attributes to Isbell (the name is uncertain in the recording; 2:03:54). In the session itself he and the room arrive at lattices, and a large ordinal; $\mathbf{Set}$ with $I = \{2\}$ is another counterexample, faithfulness holding because every set is a limit of powers of $2^{(\mathrm{e})}$. His hope of a bound on the number of limit operations (8:08:14) he retracted himself on 10 July.

4. Accessible categories

Problem 8 (`4gynb9uTEE4`, 1:31:03)**.** *Is a small 2-colimit in $\mathbf{Cat}$ of accessible categories accessible?*

Status. No in general[(e)]. Let B be the free category on arrows $a\colon u \to v$, $b\colon v \to w$, $h\colon u \to w$, map it to the chain $0 < 1 < 2$ and to the free monoid on one generator t, sending a, b, h to t. The pushout, strict or pseudo, is the one-object category with endomorphisms $\{1, t\}$ and $t = t^2$, since $h = ba$ in the chain; the idempotent t does not split, while accessible categories split idempotents. The three categories are small and idempotent-complete, hence accessible, and the functors are accessible. He expected the answer yes, while noting that a sum of two categories with colimits has none. Directed colimits of accessible categories along accessible full embeddings are accessible (Paré–Rosický [25]). The limit side, which he recalls from SGA 4, is right in modern form [20, ch. 5].

Problem 9 (`4gynb9uTEE4`, 1:51:31)**.** *Is a full subcategory of an accessible category, stable under limits and with a left adjoint inclusion, accessible?*

Status. Yes under an extra hypothesis; in ZFC alone we have not found it settled. When the inclusion is also accessible, the condition he proposed adding, the answer is yes in ZFC: for a locally presentable ambient category this is [2, 2.78]. Under Vopěnka's principle the answer is yes for a locally presentable ambient category [2, ch. 6]. Under the weak Vopěnka principle every limit-closed full subcategory of a locally presentable category is reflective, and this fails in the category of graphs when the weak principle fails [3], [2, 6.22]; Wilson showed that the weak principle does not imply the full one [30]. A reflective, limit-closed, non-accessible subcategory in a model of the weak principle without the full one would show that ZFC does not decide the question; we have not found one in print.

Problem 10 (`4gynb9uTEE4`, 4:48:17)**.** *If E and E^{op} are both accessible, is E equivalent to the point?*

Status. No: the two-element chain is a counterexample, being, like its opposite, small and idempotent-complete. With small colimits in E, as he assumes for his main theorem, E and E^{op} are both locally presentable, and then E is a preorder, hence a complete lattice (Gabriel–Ulmer [11, Satz 7.13]).

5. Algebraic geometry and group schemes

Problem 11 (`vuys5SNs2eQ`, 3:31:11; see also 2:59:31)**.** *Do immersions that are not quasi-compact descend along a faithfully flat quasi-compact morphism? At 2:59:31: is descent of subschemes exact without retrocompactness? (« you asked for an open problem »).*

He believes the first but cannot prove it; his attempt by a filtered union of quasi-compact opens fails on his own example, the closed points of the line.

Status. Both are false. Open, closed and quasi-compact immersions descend along fpqc coverings, and immersions along fppf ones [29, 02L3, 02L6, 02L8, 02YM]; on the underlying sets, locally closed and retrocompact subsets descend along fpqc morphisms, and locally closed ones along fppf morphisms [28, Thms 4.1, 4.2]. The disjoint union of the $\operatorname{Spec}\mathbf{F}_p$ mapped to $\operatorname{Spec}\mathbf{Z}$ is not an immersion, but becomes one after the faithfully flat base change $\operatorname{Spec}\prod_p \mathbf{Z}_{(p)} \to \operatorname{Spec}\mathbf{Z}$; the same subscheme answers the second question. The example, base change included ($\prod_q \mathbf{Z}_{(q)}/p \cong \mathbf{F}_p$), is in a comment of August 2026 on Stacks tag 02YL, which we checked; it is not refereed, and we found no earlier published counterexample. The subscheme is the one he used in class, the closed points, taken after base change.

Problem 12 (`3ULFIl_2Wis`, 5:04:48)**.** *« The main conjecture »: a connected separable group scheme over a field is an extension of a group of finite type by an affine group, the quotient map faithfully flat and quasi-compact; equivalently a filtered limit of groups of finite type with affine transition maps.*

Status. Proved by Perrin [26], a summary of his Orsay work: every quasi-compact group scheme over a field is a filtered limit of algebraic quotients with affine kernels, with transition maps affine from some index on; he notes that the statement had been conjectured in

EGA IV. A connected group scheme over a field is quasi-compact, as Grothendieck recalls just after the statement. Perrin's counterexample with abelian varieties of growing dimension is the phenomenon Grothendieck wanted to exclude at 5:11:21, the dimensions of the abelian quotients being bounded.

Asked for a characterisation of schemes among algebraic spaces (6:39:12, a session of the algebraic geometry course recorded on this video), he answers that it would solve the representability problem, and no; there are sufficient criteria and partial results [29, 03XX, 0ADD, 03JI, 0D2W]. Smaller questions are settled. In the algebraic geometry course: whether a Noetherian spectrum, the ascending chain condition on radical ideals, is the same as the ascending chain condition on primes (`51wmmpHand0`, 7:42:27; no: an infinite product of fields has only maximal primes and an infinite, non-Noetherian spectrum$^{(e)}$; see Ohm and Pendleton for rings with Noetherian spectrum [24]); and whether a scheme can have no closed point (`Oqhr-Gt66es`, 0:10:49; yes [29, 01IY]). There he also states that the local schemes are those with a unique closed point. That holds for quasi-compact schemes, but not in general: $W \sqcup \operatorname{Spec} k$, with W a scheme without closed point, has a unique closed point and is not local$^{(e)}$. In the group-schemes course: whether a monoid scheme with a closed unit section is separated (`3ULFIl_2Wis`, 3:56:26; no: adjoin a unit, as an extra rational point, to a non-separated scheme with a rational point c and the constant multiplication $xy = c^{(e)}$). Whether quasi-coherent sheaves descend along a universal effective epimorphism (6:27:24, algebraic geometry course) we found settled only in the affine case: there a universal effective epimorphism is a pure ring map$^{(e)}$, and descent of modules holds exactly along pure maps [29, 08XA], [22].

6. Effectivity

At the end of the algebraic geometry session of 11 July, recorded on the second group-schemes video (`3ULFIl_2Wis`, 6:59:50), asked for open problems, he gives the effectivity of finite equivalence relations on a finite scheme over a field, posed in FGA in 1961, and says he suspects a negative answer. It is negative over every field, by a restriction of Raicu's relation on the plane to a local Artin scheme of length 27; that answer is the subject of a companion note [9].

7. Summary

The problems that still look open are those on algebraic theories over a fixed base (Problem 7), the equivalence of his two infinite exactness conditions with total distributivity (Problem 1), the status in ZFC alone of Problem 9, and the descent of quasi-coherent sheaves along non-affine universal effective epimorphisms. Five of the statements he expected to hold are false (Problems 1, 2, 8, 10 and 11); his programme of classifying topoi, his main conjecture and his doubt about effectivity were borne out.

Use of language models. The questions were located in the transcripts, and the literature searched, with language models (Claude Opus 5.5, Anthropic), which also wrote a first draft of this text and the arguments marked $^{(e)}$, under the author's direction. The references were opened on their pages; the arguments marked $^{(e)}$ have not been checked by a specialist.

References

[1] M. Adelman, Abelian categories over additive ones, *J. Pure Appl. Algebra* 3 (1973), 103–117, doi:10.1016/0022-4049(73)90026-1.

[2] J. Adámek, J. Rosický, *Locally Presentable and Accessible Categories*, LMS Lecture Note Ser. 189, Cambridge University Press, 1994, doi:10.1017/CBO9780511600579.

[3] J. Adámek, J. Rosický, V. Trnková, Are all limit-closed subcategories of locally presentable categories reflective?, in *Categorical Algebra and its Applications*, LNM 1348, Springer, 1988, 1–18, doi:10.1007/BFb0081345.

[4] R. Blackwell, G. M. Kelly, A. J. Power, Two-dimensional monad theory, *J. Pure Appl. Algebra* 59 (1989), 1–41, doi:10.1016/0022-4049(89)90160-6.

Problem	His stance	Status
1 presheaf topoi	suspects yes	false: Scott sheaves on $[0,1]^{(e)}$; essential subtopoi of $\widehat{C}$ [19]
2 $f^!$	feels no	false; local morphisms [16]
3 classifying topoi	programme	done for geometric theories
4 exactness properties	open in discussion	formalised [14]; semigroup case trivial$^{(e)}$
5 algebraic = representable	aim	yes for small R, no in general
6 envelopes, localisations	known in two cases	small D: yes [18, 4]
7 problem three	neither direction proved	$\Rightarrow$ yes, $\Leftarrow$ no for all limits; fixed base open
8 colimits of accessible	expects yes	no$^{(e)}$; directed, full embeddings: yes [25]
9 reflective subcategories	asks for a counterexample	yes if the inclusion is accessible, or under Vopěnka [2]; ZFC alone open to us
10 E, E^{op} accessible	guesses the point	no: the two-element chain [11]
11 descent of immersions	believes yes	no (Stacks comment, 2026)
12 structure of group schemes	main conjecture	Perrin 1976 [26]
§6 effectivity	suspects no	no, over every field [9]

[5] M. C. Bunge, Relative functor categories and categories of algebras, *J. Algebra* 11 (1969), 64–101, doi:10.1016/0021-8693(69)90102-1.

[6] O. Caramello, Lattices of theories, arXiv:0905.0299; *Theories, Sites, Toposes*, Oxford University Press, 2018.

[7] A. Carboni, R. Celia Magno, The free exact category on a left exact one, *J. Austral. Math. Soc. Ser. A* 33 (1982), 295–301, doi:10.1017/S1446788700018735.

[8] Commutator-IO, grothendieck-project, `docs/buffalo-corrections.md`: corrections found while formalising the Buffalo courses.

[9] M. Hua, A non-effective finite equivalence relation on a finite scheme over a field, after a question of Grothendieck, 2026, `https://grothendieck.commutator.io/article/grothendieck-effective.pdf`.

[10] P. Freyd, Representations in abelian categories, in *Proc. Conf. Categorical Algebra (La Jolla, 1965)*, Springer, 1966, 95–120, doi:10.1007/978-3-642-99902-4_4.

[11] P. Gabriel, F. Ulmer, *Lokal präsentierbare Kategorien*, LNM 221, Springer, 1971, doi:10.1007/BFb0059396.

[12] J. Giraud, *Cohomologie non abélienne*, Grundlehren 179, Springer, 1971, doi:10.1007/978-3-662-62103-5.

[13] M. Hakim, *Topos annelés et schémas relatifs*, Ergebnisse der Math. 64, Springer, 1972, doi:10.1007/978-3-662-59155-0.

[14] P.-A. Jacqmin, Z. Janelidze, On stability of exactness properties under the pro-completion, *Adv. Math.* 377 (2021), 107484, doi:10.1016/j.aim.2020.107484; arXiv:2002.02204.

[15] P. T. Johnstone, A. Joyal, Continuous categories and exponentiable toposes, *J. Pure Appl. Algebra* 25 (1982), 255–296, doi:10.1016/0022-4049(82)90083-4.

[16] P. T. Johnstone, I. Moerdijk, Local maps of toposes, *Proc. London Math. Soc. (3)* 58 (1989), 281–305, doi:10.1112/plms/s3-58.2.281.

[17] P. T. Johnstone, *Stone Spaces*, Cambridge Stud. Adv. Math. 3, Cambridge University Press, 1982.

[18] G. M. Kelly, A unified treatment of transfinite constructions for free algebras, free monoids, colimits, associated sheaves, and so on, *Bull. Austral. Math. Soc.* 22 (1980), 1–83, doi:10.1017/S0004972700006353.

[19] R. B. B. Lucyshyn-Wright, Totally distributive toposes, *J. Pure Appl. Algebra* 216 (2012), 2425–2431, doi:10.1016/j.jpaa.2012.03.018; arXiv:1108.4032.

[20] M. Makkai, R. Paré, *Accessible Categories: The Foundations of Categorical Model Theory*, Contemp. Math. 104, AMS, 1989, doi:10.1090/conm/104.

[21] M. Makkai, G. E. Reyes, *First Order Categorical Logic*, LNM 611, Springer, 1977, doi:10.1007/BFb0066201.

[22] B. Mesablishvili, Pure morphisms of commutative rings are effective descent morphisms for modules—a new proof, *Theory Appl. Categ.* 7 (2000), no. 3, 38–42, `http://www.tac.mta.ca/tac/volumes/7/n3/7-03abs.html`.

[23] I. Moerdijk, *Classifying Spaces and Classifying Topoi*, LNM 1616, Springer, 1995, doi:10.1007/BFb0094441.

[24] J. Ohm, R. L. Pendleton, Rings with noetherian spectrum, *Duke Math. J.* 35 (1968), 631–639, doi:10.1215/S0012-7094-68-03565-5.

[25] R. Paré, J. Rosický, Colimits of accessible categories, *Math. Proc. Cambridge Philos. Soc.* 155 (2013), 47–50, doi:10.1017/S0305004113000030; arXiv:1110.0767.
[26] D. Perrin, Approximation des schémas en groupes, quasi compacts sur un corps, *Bull. Soc. Math. France* 104 (1976), 323–335, doi:10.24033/bsmf.1830.
[27] Commutator-IO, grothendieck-project, `https://github.com/Commutator-IO/grothendieck-project`: index of results of the three courses (`docs/buffalo-*/index.tex`) and the Buffalo tab of `https://grothendieck.commutator.io`.
[28] D. Rydh, Submersions and effective descent of étale morphisms, *Bull. Soc. Math. France* 138 (2010), 181–230, doi:10.24033/bsmf.2588.
[29] The Stacks Project Authors, *The Stacks Project*, `https://stacks.math.columbia.edu`, tags as cited.
[30] T. M. Wilson, Weak Vopěnka's principle does not imply Vopěnka's principle, *Adv. Math.* 363 (2020), 106986, doi:10.1016/j.aim.2020.106986.

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